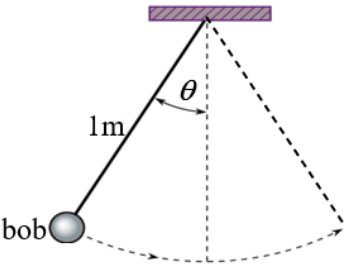


1	ACSB/2025/II/09
	Car P moves along a horizontal straight line such that, at time t seconds after passing O, its velocity v m/s, is given by $v = -3t^2 - 10t + 4$. The displacement of P from O is 28 m at the start of motion. (a) Find an expression, in terms of t, for the displacement of P. [2] Another car Q moves along the same horizontal straight line as P and starts from O at the same instant that P begins to move. The initial velocity of Q is 2 m/s and its acceleration, a m/s², at time t seconds after motion has begun, is given by $a = 2 - 6t$. (b) Find an expression, in terms of t, for the displacement of Q. [4] (c) Find the value of t at the instant when P and Q collide. [3] (d) Explain if P and Q are travelling in the same direction when the collision occurs. [3]
2	BASS/2025/I/11
	Bob drives along a straight road. His velocity, v m/s, is given by $v = 2t^2 + 3t - 5$, where t is the time in seconds after passing through O. Calculate (a) the initial acceleration, [2] (b) the value of t for which Bob is instantaneously at rest, [2] (c) the total distance that Bob travelled in the first 4 seconds. [4]
3	CCHY/2025/II/10
	A particle travels in a straight line so that its displacement, s m, from O at time t seconds, where $t \geq 0$, is modelled by $s = t^3 - \frac{9}{2}t^2 + 6t + 1$. (a) Find the values of t for which the particle is instantaneously at rest. [3] (b) The particle's acceleration is 15 m/s² at T seconds. Find the total distance travelled by the particle in the interval $t = 0$ to $t = T$. [4] (c) Explain clearly why the answer found in part (b) should not be obtained by finding the value of s when $t = T$. [2]
4	GMS/2025/II/5
	A particle P leaves a fixed point O and moves in a straight line so that, t s after leaving O, its velocity is given by $v = -3t^2 + kt + 15$, where k is a constant. When $t = 2$, its velocity is 11 ms⁻¹. (a) Show that $k = 4$. [1] (b) Calculate the value of t for which the particle is at instantaneous rest. [2] (c) Find the maximum velocity of the particle. [3] (d) Calculate the total distance travelled by the particle in the first 4 seconds. [4] (e) Find the time the particle returns to point O. [2]

5	FMSS/2025/I/12
	 A pendulum of length 1m swings back and forth such that its angular displacement θ radians from the vertical is modelled by the function $\theta(t) = a \cos bt$ where t is the time in seconds. The pendulum completes one full swing (left to right and back) in 4 seconds and the maximum angle reached is 0.5 radians. (a) Explain why $a = 0.5$ and show that $b = \frac{1}{2} \pi$. [3] (b) Obtain an expression for the velocity of the pendulum at time t and hence deduce its maximum speed in m/s. [2] (c) Find the magnitude of the acceleration of the pendulum when $\theta = 0.5$ radians after 1 cycle. [2]
6	ZHSS/2025/II/7
	A particle, moving in a straight line so that, at time t seconds after passing through a point A with a velocity of v m/s, is given by $v = 10e^{-0.1t} - 2.$ (a) Find the acceleration of the particle when it is instantaneously at rest. [5] (b) Find the expression in terms of t, for the displacement, from A, of the particle, t seconds after passing A. [3] (c) Show that the particle is again at A at some instant during the 50th second after first passing through A. [3]

1		$S_p = \int -3t^2 - 10t + 4 \, dt$ $= -t^3 - 5t^2 + 4t + c$ $t = 0, \quad S_p = 28, c = 28$ $S_p = -t^3 - 5t^2 + 4t + 28$
	(b)	$V_Q = \int 2 - 6t \, dt$ $= 2t - 3t^2 + c$ $t = 0, V_Q = 2, \quad c = 2$ $V_Q = 2t - 3t^2 + 2$ $S_Q = \int 2t - 3t^2 + 2 \, dt$ $= t^2 - t^3 + 2t + c$ $t = 0, s = 0, c = 0$ $S_Q = t^2 - t^3 + 2t$
	(c)	$S_p = S_Q$ $-t^3 - 5t^2 + 4t + 28 = t^2 - t^3 + 2t$ $6t^2 - 2t - 28 = 0$ $(3t - 7)(t + 2) = 0$ $t = 2\frac{1}{3} \text{ s} \quad \text{or } t = -2(\text{rejected})$
	(d)	$V_p = -3\left(2\frac{1}{3}\right)^2 - 10\left(2\frac{1}{3}\right) + 4 = -35\frac{2}{3} \text{ m/s}$ $V_Q = 2\left(2\frac{1}{3}\right) - 3\left(2\frac{1}{3}\right)^2 + 2 = -9\frac{2}{3} \text{ m/s}$ Since both V_p & $V_Q < 0$, P and Q are travelling in the same direction.

2	a $v = 2t^2 + 3t - 5$ $a = 4t + 3$ When $t = 0$, $a = 3$ b Instantaneously at rest $\rightarrow v = 0$ $2t^2 + 3t - 5 = 0$ $(2t + 5)(t - 1) = 0$ $t = 1 \text{ or } t = -\frac{5}{2} \text{ (rej)}$ c $s = \int 2t^2 + 3t - 5 \, dt$ $s = \frac{2t^3}{3} + \frac{3t^2}{2} - 5t + c$ When $t = 0$, $s = 0$, $c = 0$ $s = \frac{2t^3}{3} + \frac{3t^2}{2} - 5t$ When $t = 4$, $s = \frac{2(4)^3}{3} + \frac{3(4)^2}{2} - 5(4) = 46\frac{2}{3}$ When $t = 1$, $s = \frac{2(1)^3}{3} + \frac{3(1)^2}{2} - 5(1) = -2\frac{5}{6}$ Total distance = $46\frac{2}{3} + 2\left(2\frac{5}{6}\right) = 52\frac{1}{3} \text{ m}$ Alternative method $s = \int 2t^2 + 3t - 5 \, dt$ $s = \frac{2t^3}{3} + \frac{3t^2}{2} - 5t + c$ When $t = 0$, $s = 0$, $c = 0$ $s = \frac{2t^3}{3} + \frac{3t^2}{2} - 5t$ $[s]_0^1 + [s]_1^4 = 52\frac{1}{3}$
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3

(a)

$$v = \frac{ds}{dt} = 3t^2 - 9t + 6$$

When particle is at instantaneously rest, $v = 0$.

$$3t^2 - 9t + 6 = 0$$

$$t^2 - 3t + 2 = 0$$

$$(t-1)(t-2) = 0$$

$$t = 1 \text{ or } t = 2$$

(b)

$$a = \frac{dv}{dt} = 6t - 9$$

When $a = 15$, $6T - 9 = 15$

$$T = 4$$

When $t = 0$, $s = 1$

When $t = 1$, $s = 3.5$

When $t = 2$, $s = 3$

When $t = 4$, $s = 17$

$$\begin{aligned} \text{Total distance} &= (3.5 - 1) + (3.5 - 3) + (17 - 3) \\ &= 17 \end{aligned}$$

(c)

Even though the numerical value is the same, the particle turned twice (at least once), thus will need to account for the additional distance. Also, evaluating s at $t = T$ only gave the distance of the particle from O , which is not what is required.

4

(a)

$$11 = -3(2)^2 + k(2) + 15$$

$$k = 4$$

(b)

At rest, $v = 0$

$$-3t^2 + 4t + 15 = 0$$

$$3t^2 - 4t - 15 = 0$$

$$(3t+5)(t-3) = 0$$

$$t = -\frac{5}{3} \text{ (NA)} \quad \text{or} \quad t = 3$$

(c)

$$v = -3t^2 + 4t + 15$$

$$a = -6t + 4$$

When velocity is maximum, $a = 0$

$$t = \frac{2}{3}$$

$$\begin{aligned} \text{Maximum velocity} &= -3\left(\frac{2}{3}\right)^2 + 4\left(\frac{2}{3}\right) + 15 \\ &= 16\frac{1}{3} \end{aligned}$$

(d)

$$s = \int (-3t^2 + 4t + 15) dt$$

$$= -t^3 + 2t^2 + 15t + c$$

When $t = 0$, $s = 0$, $\therefore c = 0$

$$s = -t^3 + 2t^2 + 15t$$

$$\begin{aligned} \text{When } t = 3, \quad s &= -3^3 + 2(3)^2 + 15(3) \\ &= 36 \end{aligned}$$

$$\begin{aligned} \text{When } t = 4, \quad s &= -4^3 + 2(4)^2 + 15(4) \\ &= 28 \end{aligned}$$

$$\begin{aligned} \text{Distance travelled in first 4 seconds} &= 36 + (36 - 28) \\ &= 44 \end{aligned}$$

(e)

At point O , $s = 0$

$$-t^3 + 2t^2 + 15t = 0$$

$$-t(t^2 - 2t - 15) = 0$$

$$-t(t - 5)(t + 3) = 0$$

$$t = 0 \text{ (NA)}, \quad t = -3 \text{ (NA)}, \quad t = 5$$

5

(a) Amplitude, a , refers to the maximum angle/displacement of the pendulum swing from the vertical. So $a = 0.5$ (shown)

$$\text{Period} = \frac{2\pi}{b} = 4$$

$$b = \frac{2\pi}{4}$$

$$= \frac{1}{2}\pi \text{ (shown)}$$

(b)

$$\theta(t) = 0.5 \cos \frac{\pi}{2} t$$

$$v = -\frac{\pi}{4} \sin \frac{\pi}{2} t$$

$$\text{Maximum speed} = \frac{\pi}{4} \text{ m/s or } 0.785 \text{ m/s (to 3 s.f.)}$$

(c)

$$a = \frac{dv}{dt} = -\frac{\pi^2}{8} \cos \frac{\pi}{2} t$$

When $\theta = 0.5$ radians after 1 cycle, $t = 4$,

$$a = -\frac{\pi^2}{8} \cos \frac{\pi}{2} (4)$$

$$= -\frac{\pi^2}{8} \cos 2\pi$$

$$= -\frac{\pi^2}{8}$$

$$\text{Magnitude of acceleration is } \frac{\pi^2}{8} \text{ m/s}^2 \text{ or } 1.23 \text{ m/s}^2 \text{ (to 3 s.f.)}$$

6**(a)**At instantaneous rest, $v = 0$

$$10e^{-0.1t} - 2 = 0$$

$$e^{-0.1t} = \frac{1}{5}$$

$$-0.1t = \ln \frac{1}{5}$$

$$t = -10 \ln \frac{1}{5} \text{ s or } 16.094 \text{ s (5 sig fig)}$$

$$a = \frac{dv}{dt} = 10(-0.1)e^{-0.1t}$$

$$= -e^{-0.1t}$$

$$t = -10 \ln \frac{1}{5}$$

$$a = -e^{-0.1(-10 \ln \frac{1}{5})}$$

$$= -0.2 \text{ m/s}^2 \text{ or } -0.200 \text{ m/s}^2 \text{ (3 sig fig)}$$

(b)

$$s = \int 10e^{-0.1t} - 2 \, dt$$

$$= \frac{10e^{-0.1t}}{-0.1} - 2t + c$$

$$= -100e^{-0.1t} - 2t + c$$

when $t = 0, s = 0$

$$c = 100$$

$$s = -100e^{-0.1t} - 2t + 100$$

(c)

$$t = 49$$

$$s = -100e^{-0.1(49)} - 2(49) + 100$$

$$= 1.255$$

$$t = 50$$

$$s = -100e^{-0.1(50)} - 2(50) + 100$$

$$= -0.674$$

Since the displacement changes from a positive value when $t = 49$ to a negative value when $t = 50$, the particle passed through point A during the 50th sec.